

Process-to-Process Delivery

- The data link layer is responsible for delivery of frames between two neighboring nodes over a link. This is called *node-to-node delivery*.
- The network layer is responsible for delivery of datagrams between two hosts. This is called *host-to-host delivery*.
- Communication on the Internet is not defined as the exchange of data between two nodes or between two hosts. Real communication takes place between two processes (application programs).
- We need process-to-process delivery. However, at any moment, several processes may be running on the source host and several on the destination host. To complete the delivery, we need a mechanism to deliver data from one of these processes running on the source host to the corresponding process running on the destination host.
- The transport layer is responsible for process-to-process delivery-the delivery of a packet, part of a message, from one process to another. Two processes communicate in a client/server relationship.

Addressing

- Whenever we need to deliver something to one specific destination among many, we need an address. At the data link layer, we need a MAC address to choose one node among several nodes if the connection is not point-to-point.
- A frame in the data link layer needs a destination MAC address for delivery and a source address for the next node's reply.
- At the network layer, we need an IP address to choose one host among millions. A datagram in the network layer needs a destination IP address for delivery and a source IP address for the destination's reply.
- At the transport layer, we need a transport layer address, called a port number, to choose among multiple processes running on the destination host.

- The destination port number is needed for delivery; the source port number is needed for the reply.

IANA Ranges

The IANA (Internet Assigned Number Authority) has divided the port numbers into three ranges: well known, registered, and dynamic (or private),

- Well-known ports. The ports ranging from 0 to 1023 are assigned and controlled by IANA. These are the well-known ports.
- Registered ports. The ports ranging from 1024 to 49,151 are not assigned or controlled by IANA. They can only be registered with IANA to prevent duplication.
- Dynamic ports. The ports ranging from 49,152 to 65,535 are neither controlled nor registered. They can be used by any process. These are the ephemeral ports.

Socket Addresses

- Process-to-process delivery needs two identifiers, IP address and the port number, at each end to make a connection. The combination of an IP address and a port number is called a socket address. The client socket address defines the client process uniquely just as the server socket address defines the server process uniquely
- A transport layer protocol needs a pair of socket addresses: the client socket address and the server socket address. These four pieces of information are part of the IP header and the transport layer protocol header.
- The IP header contains the IP addresses; the UDP or TCP header contains the port numbers.

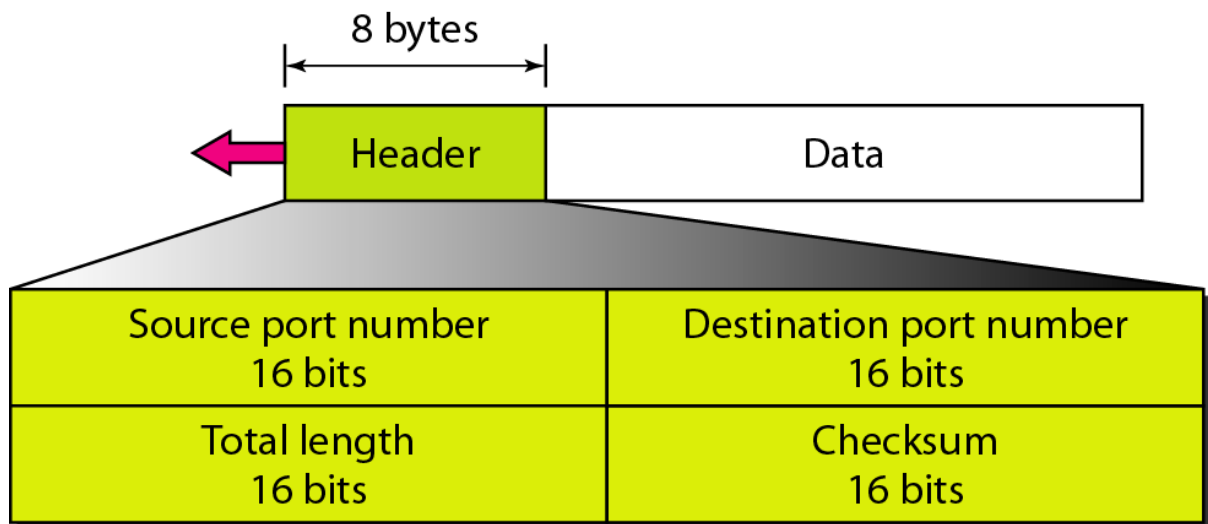
Connectionless Service

- In a connectionless service, the packets are sent from one party to another with no need for connection establishment or connection release. The packets are not numbered; they may be delayed or lost or may arrive out of sequence. There is no acknowledgment either.
- UDP, is connectionless.

Connection-Oriented Service

- In a connection-oriented service, a connection is first established between the sender and the receiver. Data are transferred. At the end, the connection is released.
- TCP and SCTP are connection-oriented protocols.

User Datagram Protocol



Source port number:

- This is the port number used by the process running on the source host. It is 16 bits long, which means that the port number can range from 0 to 65,535.
- If the source host is the client (a client sending a request), the port number, in most cases, is an ephemeral port number requested by the process and chosen by the UDP software running on the source host.
- If the source host is the server (a server sending a response), the port number, in most cases, is a well-known port number.

Destination port number:

- This is the port number used by the process running on the destination host. It is also 16 bits long.

- If the destination host is the server (a client sending a request), the port number, in most cases, is a well-known port number.
- If the destination host is the client (a server sending a response), the port number, in most cases, is an ephemeral port number. In this case, the server copies the ephemeral port number it has received in the request packet.

Length:

- This is a 16-bit field that defines the total length of the user datagram, header plus data. The 16 bits can define a total length of 0 to 65,535 bytes.
- However, the total length needs to be much less because a UDP user datagram is stored in an IP datagram with a total length of 65,535 bytes.
- The length field in a UDP user datagram is actually not necessary. A user datagram is encapsulated in an IP datagram.
- There is a field in the IP datagram that defines the total length. There is another field in the IP datagram that defines the length of the header. So if we subtract the value of the second field from the first, we can deduce the length of a UDP datagram that is encapsulated in an IP datagram.

Checksum:

- This field is used to detect errors over the entire user datagram (header plus data).

Uses of UDP

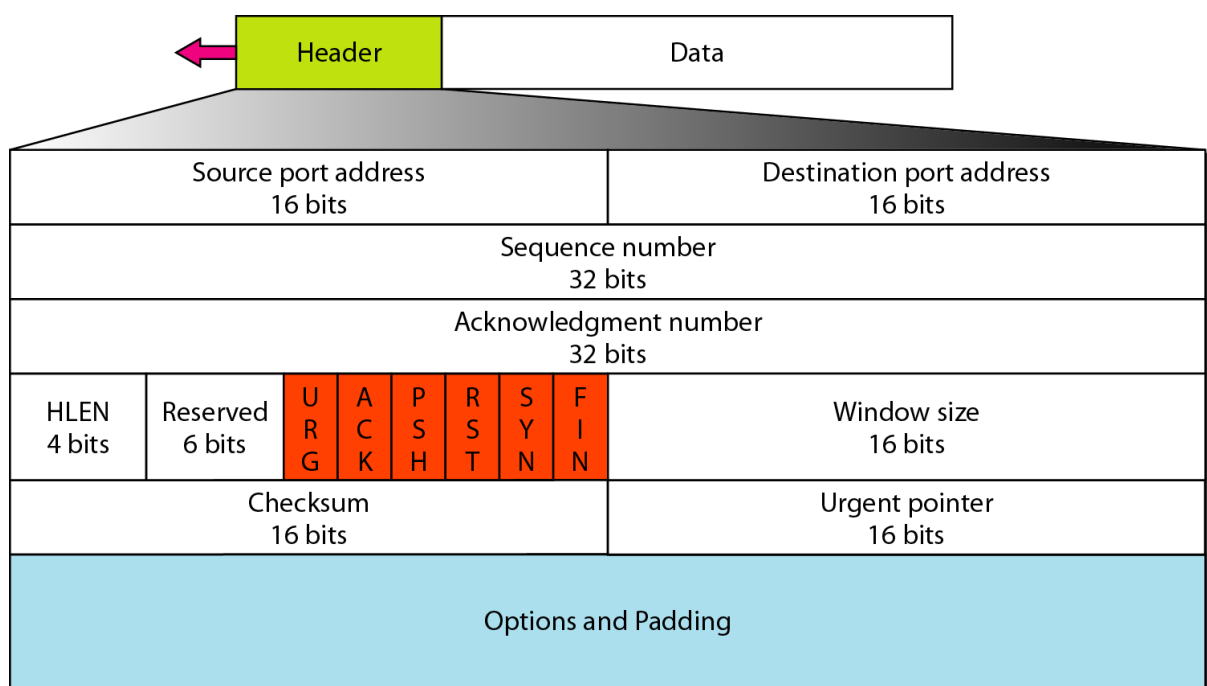
The following lists some uses of the UDP protocol:

- UDP is suitable for a process that requires simple request response communication with little concern for flow and error control. It is not usually used for a process such as FrP that needs to send bulk data.
- UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer

Protocol (TFTP) process includes flow and error control. It can easily use UDP.

- UDP is a suitable transport protocol for multicasting. Multicasting capability is embedded in the UDP software but not in the TCP software.
- UDP is used for management processes such as SNMP.
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP)

TCP



- The segment consists of a 20- to 60-byte header, followed by data from the application program.
- The header is 20 bytes if there are no options and up to 60 bytes if it contains options.

Source port address:

- This is a 16-bit field that defines the port number of the application program in the host that is sending the segment. This serves the same purpose as the source port address in the UDP header.

Destination port address:

This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment. This serves the same purpose as the destination port address in the UDP header.

Sequence number:

- This 32-bit field defines the number assigned to the first byte of data contained in this segment. TCP is a stream transport protocol.
- To ensure connectivity, each byte to be transmitted is numbered. The sequence number tells the destination which byte in this sequence comprises the first byte in the segment.
- During connection establishment, each party uses a random number generator to create an initial sequence number (ISN), which is usually different in each direction.

Acknowledgment number:

- This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the other party. If the receiver of the segment has successfully received byte number x from the other party, it defines $x + 1$ as the acknowledgment number.
- Acknowledgment and data can be piggybacked together.

Header length:

- This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes. Therefore, the value of this field can be between 5 ($5 \times 4 = 20$) and 15 ($15 \times 4 = 60$).

Reserved:

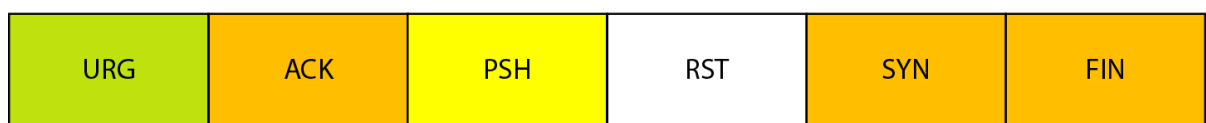
This is a 6-bit field reserved for future use.

Control:

- This field defines 6 different control bits or flags
- One or more of these bits can be set at a time.

URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push

RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection



Types of Flags:

Synchronization (SYN) – It is used in first step of connection establishment phase or 3-way handshake process between the two hosts. Only the first packet from sender as well as receiver should have this flag set. This is used for synchronizing sequence number i.e. to tell the other end which sequence number they should expect.

Acknowledgement (ACK) – It is used to acknowledge packets which are successful received by the host. The flag is set if the acknowledgement number field contains a valid acknowledgement number.

In given below diagram, the receiver sends an ACK = 1 as well as SYN = 1 in the second step of connection establishment to tell sender that it received its initial packet.

Finish (FIN) – It is used to request for connection termination i.e. when there is no more data from the sender, it requests for connection termination. This is the last packet sent by sender. It frees the reserved resources and gracefully terminate the connection.

Reset (RST) – It is used to terminate the connection if the RST sender feels something is wrong with the TCP connection or that the conversation should not exist. It can get send from receiver side when packet is send to particular host that was not expecting it.

Push (PSH) – Transport layer by default waits for some time for application layer to send enough data equal to maximum segment size so that the number of packets transmitted on network minimizes which is not desirable by some application like interactive applications(chatting).

- Similarly transport layer at receiver end buffers packets and transmit to application layer if it meets certain criteria.

- This problem is solved by using PSH. Transport layer sets PSH = 1 and immediately sends the segment to network layer as soon as it receives signal from application layer. Receiver transport layer, on seeing PSH = 1 immediately forwards the data to application layer. In general, it tells the receiver to process these packets as they are received instead of buffering them.

Urgent (URG) –Data inside a segment with URG = 1 flag is forwarded to application layer immediately even if there are more data to be given to application layer. It is used to notify the receiver to process the urgent packets before processing all other packets. The receiver will be notified when all known urgent data has been received.

Window size:

- This field defines the size of the window, in bytes, that the other party must maintain. Note that the length of this field is 16 bits, which means that the maximum size of the window is 65,535 bytes. This value is normally referred to as the receiving window (rwnd) and is determined by the receiver.
- The sender must obey the dictation of the receiver in this case.

Checksum:

This 16-bit field contains the checksum. The calculation of the checksum for TCP follows the same procedure as the one described for UDP. However, the inclusion of the checksum in the UDP datagram is optional, whereas the inclusion of the checksum for TCP is mandatory.

Urgent pointer:

- This 16-bit field, which is valid only if the urgent flag is set, is used when the segment contains urgent data.
- It defines the number that must be added to the sequence number to obtain the number of the last urgent byte in the data section of the segment.

Options:

- There can be up to 40 bytes of optional information in the TCP header.